Chris Thomas Saji

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Summary

AI/Software Engineer and Gen AI developer with an M.Tech in Artificial Intelligence and Machine Learning, building production computer-vision systems, scalable Spring Boot microservices and LLM-powered products, with a high level of experience shipping full-stack software through Claude Code.

Skills

LANGUAGES	Python, Java, C++, SQL, TypeScript
GENERATIVE AI	Gen AI developer: LLM apps and AI agents (Claude, OpenAI and Gemini APIs, LangGraph), structured outputs, multi-model failover, evals and LLM cost tracking
CLAUDE CODE	High level of experience: agentic development with subagents, custom skills, hooks, MCP servers and parallel sessions in git worktrees, gated by automated tests
AI/ML	Computer Vision, NLP, Deep Learning; InsightFace (SCRFD/ArcFace), FAISS, CLIP, ONNX Runtime, PyTorch, TensorFlow, OpenCV
BACKEND	Spring Boot, Spring Cloud Gateway, FastAPI, Celery, Microservices (CQRS/DCI), RabbitMQ, PostgreSQL, Redis, REST, JWT/OAuth2
FRONTEND & DESKTOP	Next.js, React, Electron, Redux, Vite
CLOUD & DEVOPS	Docker, GitHub Actions, Terraform (GCP), Cloudflare Tunnel, Sentry
TOOLS	Git/GitHub, Maven, VS Code, Jupyter, Google Colab

Experience

Mastermine Technologies – Kochi, Kerala **Aug 2025 – Present**
AI/Software Engineer

- Build the face-recognition pipeline for ClickMomentz, a live AI photo platform for large-scale events, using InsightFace SCRFD detection and ArcFace 512-d embeddings on ONNX Runtime with FAISS cosine-similarity indexing: guests find their photos from a single selfie at about 95% matching accuracy, with near-instant vector search; collaborate in a 6-engineer team.
- Develop a multi-stage AI culling engine scoring blur, exposure, eye-open, pose and aesthetics with CLIP scene understanding and ArcFace/HDBSCAN face clustering, plus burst and duplicate detection, to auto-rank events of 2,000 to 5,000 photos (up to 10,000); photographers set how many to keep, saving hours of manual culling per event.
- Architect 4 Spring Boot 3.5 / Java 21 microservices on a DCI + CQRS design with PostgreSQL, OpenFeign, Resilience4j circuit breakers and Caffeine caching.
- Design 8+ RabbitMQ queues for resumable async workflows (face indexing, photo matching, selfie enhancement, album archiving) with batch multiprocessing across parallel workers, indexing a typical event in under an hour.
- Implement multi-tenant, org-scoped data with JWT/OAuth2 security, a subscription/entitlements service, and S3/GCS storage with presigned URLs.
- Contribute to the cross-platform Electron 29 + React 19 + TypeScript desktop app (Redux, local SQLite) that bundles the Python AI worker for fast offline processing.

OpenText – Bengaluru **Aug 2023 – Feb 2024**
DevOps Intern

- Performed manual and Selenium-automated testing, building and maintaining test cases and automation frameworks; collaborated with developers to resolve issues and improve product quality.

Education

Rajagiri School of Engineering and Technology Kochi, Kerala | May 2026
Master of Technology ***(M.Tech.)*** *in* ***Artificial Intelligence and Machine Learning*** | *CGPA 8.41*

Jain University (NIRF Rank 65) Bengaluru | Jun 2023
Bachelor of Technology ***(B.Tech.)*** *in* ***Computer Science & Technology*** *(AI specialization)* | *CGPA 8.02*

Projects

FlowPrep: AI Tutor for JEE and NEET flowprep.in
Next.js 16, TypeScript, PostgreSQL (Supabase), Drizzle, Inngest, LangGraph, Claude and OpenAI APIs, Claude Code

- An AI tutor for JEE and NEET: lessons across 99 chapters with 300+ interactive diagrams, an AI tutor inside lessons and practice, adaptive practice, revision with a mistake notebook, and a weekly progress report.
- Co-lead a two-person team and own the platform, product surfaces and payments: dashboard, reports, settings, onboarding with parental consent for minors, Razorpay billing, PWA and GitHub Actions pipelines.
- Built entirely with Claude Code: parallel agent sessions in git worktrees, custom skills, hooks and subagent workflows, steered by architecture docs and project rules and gated by about 15,000 automated tests.
- AI gateway with Claude as primary and OpenAI as fallback, a LangGraph tutor with its checkpoints in Postgres, and background jobs that keep model calls off the request path.
- Shipped the pre-launch site on Hetzner (Docker Compose, Caddy, Cloudflare Tunnel) with a private beta sign-up.

ProofML: AI Proofreading for Malayalam Publishers proofml.streamforge.co.in
Python, FastAPI, Celery, PostgreSQL, Redis, React, PyMuPDF, Gemini/OpenAI/Claude APIs, Docker, Claude Code

- Live AI proofreading platform for Malayalam print publishers (English and Hindi too), used by a leading educational publisher and individual users: editors upload a PDF or Word file, review AI-flagged errors in the browser, fix them and download the result.
- Two-track detection: rule-based morphology and spell checking plus an LLM scan with automatic failover across Gemini, OpenAI and Claude, held at 100% recall on a curated evaluation set.
- Decodes legacy non-Unicode Malayalam fonts into Unicode, and writes editors' corrections back into the PDF in the publication's own embedded font, each one undoable.
- Multi-tenant SaaS built with Claude Code: Clerk auth, organisation isolation, metered monthly plans and a per-call LLM cost ledger.
- Led the React frontend and cloud platform (Terraform for GCP, GitHub Actions); runs on Docker Compose behind a Cloudflare Tunnel.

Vehicle Re-Identification with Temporal Consistency Module M.Tech thesis
Python, PyTorch, TransReID

- Enhanced the TransReID transformer for vehicle re-identification with a physics-aware Temporal Consistency Module (TCM) that uses camera-network topology and travel-time estimation to prune impossible matches.
- Trained and validated on the VeRi-776 benchmark, improving both matching accuracy and computational efficiency over the TransReID baseline, in a design aimed at multi-camera smart-surveillance systems.